CASE

Daniel, 35 · sudden **flank pain radiating to the groin** · cannot lie still · gross hematuria · CT: **4-mm stone in the mid-ureter** with proximal hydronephrosis · **nephrolithiasis**

BIO 004 · WEEK 6 · CASE DEEP DIVE

The Urinary System

A 35-year-old in agony from a 4-millimeter stone wedged in his ureter. The pain map is anatomy.

PATIENT CASE

Daniel is 35. He was at his desk when he developed **sudden, severe pain in his right flank** that quickly migrated forward and down toward his right groin. He cannot find a comfortable position, alternately pacing and writhing.

In the ED his pain is 10/10. Urine dipstick shows **gross hematuria**. Non-contrast CT reveals a **4-mm calcified stone in the mid-right ureter** with **proximal hydronephrosis** (dilation of the renal pelvis and proximal ureter). The diagnosis is **obstructing right ureteral stone**. He is given IV fluids, analgesia, and an alpha-blocker to help the stone pass.

A 4-millimeter stone — smaller than a pencil eraser — produces some of the worst pain in medicine. The pain map tells you where the stone is. Understanding why requires walking the urinary tract from glomerulus to urethra.

> GUIDING QUESTION
>
> How is a kidney built, what is a nephron, how does urine travel from there to the outside, and why does a stone wedged in the ureter produce pain that radiates from flank to groin?

PREWORK

Companion to your Week 6 Urinary System prework page. Terms, video, and spaced recall for the kidneys, nephron, ureters, bladder, and urethra live there.

1 KIDNEYS, GROSS

Two bean-shaped organs in the retroperitoneum

The kidneys sit **retroperitoneal** between T12 and L3, the right slightly lower than the left (pushed down by the liver). Each weighs ~150 g. Surrounded by adipose tissue (perirenal fat) and fascia (renal fascia) that anchor them.

- **Cortex**
  Outer layer where renal corpuscles live.
- **Medulla**
  Inner region of cone-shaped *renal pyramids* separated by columns of cortical tissue.
- **Minor calyces → major calyces → renal pelvis**
  Funnel system that collects urine from the papillae of the pyramids.
- **Renal hilum**
  Medial concavity where renal artery enters, renal vein exits, and ureter exits.

2 NEPHRON

The functional unit, ~1 million per kidney

Each nephron has two parts: a **renal corpuscle** (filter) and a **renal tubule** (modifier).

Glomerulus
Tuft of fenestrated capillaries inside Bowman's capsule. Pressure-driven filtration of plasma → glomerular filtrate.

Bowman's capsule
Double-walled cup surrounding glomerulus. Inner layer of *podocytes* has pedicels that interdigitate to form filtration slits.

Proximal convoluted tubule (PCT)
Most reabsorption (glucose, amino acids, ~65% of Na+ and water).

Loop of Henle
Descending limb (water permeable) + ascending limb (NaCl reabsorption). Sets up the medullary concentration gradient.

Distal convoluted tubule (DCT)
Fine-tunes Na+ and Ca++ under aldosterone and PTH.

Collecting duct
Final water reabsorption under ADH control. Multiple nephrons drain into one duct → papillary duct → minor calyx.

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3 FILTRATION → URINE

Three processes, in order

- **Filtration**: glomerular capillaries leak plasma (minus most proteins and cells) into Bowman's capsule. ~180 L/day.
- **Reabsorption**: tubule cells return water, glucose, amino acids, ions back to peritubular capillaries. ~99% of filtrate is reabsorbed.
- **Secretion**: tubule cells actively transport additional substances (H+, K+, drugs, toxins) from blood into filtrate.

Final urine output ~1-2 L/day, highly variable with hydration. Composition reflects what the body needs to dump (urea from protein breakdown, creatinine from muscle, excess electrolytes).

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4 URETERS

Muscular tubes that move urine by peristalsis

The **ureters** (25-30 cm) carry urine from renal pelvis to bladder. Walls have three layers: transitional epithelium (urothelium) inside, smooth muscle in the middle, fibrous adventitia outside. They are retroperitoneal, cross over the iliac vessels, and enter the bladder at an oblique angle (which prevents backflow).

> **Three narrowings where stones lodge.** Stones get stuck at the three anatomical constrictions: (1) *ureteropelvic junction* (where pelvis becomes ureter), (2) *pelvic brim* (where ureter crosses the iliac vessels — **Daniel's stone**), and (3) *ureterovesical junction* (where ureter enters bladder). Smaller stones (<5 mm) often pass spontaneously; larger ones may need intervention.

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5 BLADDER

Reservoir with a clever design

The **urinary bladder** sits behind the pubic symphysis. Three layers: *transitional epithelium* (allows stretch), *detrusor muscle* (smooth muscle, contracts during voiding), *adventitia*. The mucosa has wrinkles (*rugae*) that flatten as the bladder fills.

The **trigone** is a smooth triangular area on the posterior bladder floor with three corners: the two ureteral openings (above) and the urethral opening (below). The trigone is the most sensitive part of the bladder and the most common site for bladder cancer.

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6 URETHRA

Final pathway out, with male/female differences

FEATURE	FEMALE	MALE
Length	~4 cm	~20 cm
Function	urine only	urine AND semen
Regions	single	prostatic, membranous, spongy
Sphincters	internal (smooth, involuntary) + external (skeletal, voluntary at urogenital diaphragm)	same plus surrounding pelvic floor muscles

The shorter female urethra is why women have higher rates of UTIs — bacteria from the perineum reach the bladder more easily.

✎ **DRAW ZONE 1** -- *Sketch a kidney in coronal section. Label cortex, medulla, pyramids, calyces, pelvis, hilum. Then draw a nephron with all named parts (corpuscle, PCT, loop of Henle, DCT, collecting duct).*

Star the cortex (PCT, DCT, glomerulus) vs medulla (loop of Henle, collecting ducts).

✎ **DRAW ZONE 2** -- *Diagram the urinary tract from kidney to urethra. Mark the three ureteral narrowings where stones lodge. Show Daniel's stone at the iliac vessel crossing.*

Color the hydronephrosis (dilated pelvis and proximal ureter) proximal to the stone.

RETURNING TO DANIEL

Why the pain moves and why it's so severe

Daniel's pain is referred from a smooth-muscle tube spasming around an obstruction. The path the pain takes traces the embryonic origin of the ureter.

"*Pain that radiates from flank to groin*"

Visceral pain from the upper ureter refers to dermatomes T10-T12 (flank); lower ureter refers to L1-L2 (groin and inner thigh). Stone migration down the ureter changes the dermatome of referred pain.

"*Why he can't lie still*"

Renal colic patients are classically restless (versus peritonitis patients, who lie very still). The pain is not relieved by position. Spasm of the ureteral smooth muscle is rhythmic and severe.

"*Gross hematuria*"

The stone abrades the ureteral mucosa as it migrates and as the ureter contracts against it. Microscopic hematuria is nearly universal; gross hematuria is common.

"*Proximal hydronephrosis*"

Urine can't pass the obstruction. Backs up into the renal pelvis and calyces, distending them. Prolonged severe obstruction can damage the kidney (days to weeks).

"*Why an alpha-blocker*"

Tamsulosin relaxes ureteral smooth muscle, especially in the distal ureter, helping the stone pass.

"*Composition of stones*"

~80% calcium oxalate. Others: uric acid, struvite (infection-associated), cystine. Composition guides prevention (hydration, dietary changes, sometimes medications).

Kidney, nephron, ureter, bladder. Daniel's pain map is the anatomy of the urinary tract being read backwards through his dermatomes.